

PLANETARY SR III — MEXICO

Formal Capacity, Informal Continuity, Territorial Inequality, and the Human Buffer

SignalRupture (SR) | Planetary Country Study | Evidence window 2000–2025 | 2026

Scope and Evidentiary Position

This study treats the country as a coupled institutional–population system rather than a single actor or a scalar condition. The empirical window is 2000–2025 wherever comparable series permit, with later publication dates used only to recover measurements whose reference periods fall inside that window. Pressure, continuity, drift, collapse, and reorganization are evaluated by function, population, place, and duration. A deteriorating indicator is not collapse; a rising input is not capacity; and a compensatory mechanism is not presumed harmful. Official statistics, administrative records, regulatory materials, audited institutional evidence, and peer-reviewed research receive priority. Observations, SR-derived calculations, mechanism interpretations, and prospective predictions remain separately labeled. No scalar national-collapse score is constructed.

Mexico is examined as a federal republic with globally integrated production, extensive formal institutions, deep territorial inequality, high labour informality, large unpaid and household economies, migration and remittance circuits, Indigenous and communal governance, and uneven state presence. Its central empirical configuration is dual-system continuity: formal systems and informal, household, community, and transnational infrastructures jointly supply function. Informality is not treated as failure by definition, and community provision is not treated as proof of state absence.

Abstract

Mexico combines major industrial, public, fiscal, cultural, demographic, and community capacities with persistent territorial inequality and extensive informal provision. The central national pattern is dual-system continuity. Formal employment, health, education, justice, water, energy, finance, and administrative systems operate alongside household work, informal markets, kinship, migration, remittances, communal authority, and community security. This architecture can diversify suppliers and maintain function where formal reach is incomplete. It can also transfer cost and risk to women, informal workers, migrants, families, small municipalities, Indigenous communities, and populations exposed to violence or climate hazards.

The study applies the SignalRupture field system, Pressure Ecology theorem, cross-field bridge protocol, dependency topology, and Structural Restoration Dynamics across labour, income, social protection, housing, water, food, healthcare, education, childhood, care, governance, security, justice, demography, migration, digital systems, media, energy, transport, industry, climate, ecology, and disaster capacity. Official evidence supplies decisive countercurrents. Multidimensional poverty fell from 43.2% in 2016 to 29.6% in 2024, or 38.5 million people; extreme poverty also declined. Between 2022 and 2024, deprivation in access to health services declined from 39.1% to 34.2%, while 48.2% still lacked access to social security in 2024. Internet use reached 83.1% of people aged six or older in 2024, with a persistent urban–rural gap. Labour informality remained near 54% in 2024, and 33,241 homicides were registered. These observations reject national collapse while confirming unequal effective access and the continuing importance of non-state and transnational buffers.

The strongest bridge hypotheses connect informality to social protection and household care; migration and remittances to consumption, housing, and local continuity; water and climate exposure to health, schooling, and mobility; violence to business, journalism, governance, and forced movement; and digital expansion to administrative access, markets, media, and new dependency. The integrated causal sequence remains incompletely identified because formal and informal functions are poorly linked in common panels and because under-registration can be endogenous to pressure. The conditional 2031 forecast expects selective buffer saturation, not synchronized national failure: burden should become visible first among caregivers, informal workers, remittance-dependent households, health and education workers, water-insecure communities, journalists, defenders, and low-capacity municipalities.

The country model also registers the transition from computational recognition infrastructure (CRI) to AI-mediated institutional substrates (AIS) and multi-agent coordination systems (MACS). Mexico is modeled simultaneously as a major receiver of U.S.-centered pressure and as a transmission state whose manufacturing, labour, migration, remittance, ecological, and digital relations can move pressure onward. Developmental lock-in is treated as a conditional, falsifiable pathway rather than an inevitable national fate.

1. Research Object and National Question

The national question is whether the frozen SR grammar identifies a distinct, measurable Mexican configuration without converting poverty, informality, violence, or weak access into rhetorical collapse. Mexico is a hard test because

formal and informal systems are not separate worlds. They are mutually constitutive: formal firms use informal supply and care systems; households combine wages, self-employment, transfers, remittances, and public programs; public institutions rely on family navigation and unpaid support; communities govern resources where state capacity is partial.

Pressure → Institutional Reflex → Drift → Collapse → Reorganization

The sequence is evaluated by domain and territory. Collapse requires a sustained loss of essential function for a defined population after compensation is exhausted. High burden alone is pressure. Informality alone is not collapse. Community provision alone is not state absence. Reorganization requires a durable change in supplier, financing, geography, authority, or mode of delivery.

2. Canonical Definitions

Construct	Operational meaning in Mexico	Guardrail
Durable capacity	Reliable formal or community capacity that can be renewed without extraordinary sacrifice.	Physical assets are not counted as accessible capacity without affordability, staffing and security.
Effective access	Usable function after geography, price, eligibility, language, documentation, safety and reliability.	Formal entitlement is not assumed to equal use.
Compensation	Extra work, unpaid care, borrowing, remittances, self-provision, migration, private purchase or informal coordination.	Compensation may be chosen and valuable; burden must be measured.
Human Recovery Slack	Remaining time, health, income, trust, staffing and relational capacity after ordinary obligations.	Not falsely calibrated as a national scalar.
Collapse	Sustained essential-function loss after compensatory capacity crosses a registered threshold.	Poverty, homicide or informality alone cannot establish it.
Reorganization	Durable transfer of function or authority across state, market, household, community or coercive systems.	Temporary emergency response is insufficient.

3. Frozen Hypotheses and Falsifiers

Hypothesis	Mexican prediction	Failure condition
H1 Dual-system continuity	Formal output depends partly on informal, household and community capacity.	Formal and relational systems are empirically independent.
H2 Human-buffer precedence	Human Recovery Slack deteriorates before later visible functional loss.	Function consistently fails first or buffers replenish.
H3 Surface–substrate divergence	National growth or coverage can improve while effective access remains territorial and stratified.	Direct-condition measures converge with headline measures.
H4 Territorial propagation	Water, security, health, transport, education and livelihood pressures transfer locally.	No reproducible cross-domain lag remains after controls.
H5 Reorganization	Persistent gaps change suppliers toward households, remittances, informal markets or local governance.	Substitution is trivial or temporary.
H6 Incremental value	Frozen SR variables improve warning beyond mature poverty, informality and violence models.	Strong conventional baselines match or outperform SR.

4. Method and Evidence Architecture

The evidence hierarchy prioritizes INEGI, CONEVAL, Banco de México, federal administrative statistics, public-health surveillance, CFE and energy regulators, education authorities, international organizations using official Mexican series, and peer-reviewed research. National aggregates are never treated as observations of the same people. Cross-domain recurrence identifies candidate mechanisms; causal claims require linked panels or credible designs.

Effective capacity is represented structurally as $D_{\text{eff}} = D \times A \times \text{Aff} \times \text{Elig} \times \text{Safe} \times \text{Rel}$, where availability, affordability, eligibility, safety and reliability convert gross capacity into usable function. The expression is a bookkeeping architecture rather than a universal estimated product.

5. Historical and Institutional Formation

Mexico's contemporary system carries the interaction of Indigenous polities, colonial extraction, land concentration, revolution, corporatist state-building, import substitution, debt crisis, liberalization, democratization, decentralization, migration, and North American production integration. These layers produced both durable state capacities and uneven territorial incorporation. Municipalities, ejidos, communal authorities, unions, churches, families, firms, criminal organizations and civil society may all supply governance functions, but with radically different legitimacy and accountability.

Federalism distributes authority across the federation, 32 federal entities and more than 2,400 municipalities. Local fiscal and administrative capacity varies widely. The same national right can therefore convert into different lived access depending on state institutions, municipal resources, rurality, Indigenous language, security and transport. Planetary SR treats this variation as part of the mechanism rather than residual noise.

6. Political Economy, Production, and Distribution

Mexico combines advanced export manufacturing, energy and mineral resources, agriculture, tourism, financial integration and large domestic markets with low productivity in many firms and persistent informality. Nearshoring and supply-chain integration create investment opportunities, but benefits depend on electricity, water, logistics, security, housing, skills and local governance. Aggregate industrial expansion can therefore create new pressures in the places that host it.

The central distribution question is whether productivity and wage gains create durable household slack or are absorbed by housing, food, transport, care and insecurity. The reduction in multidimensional poverty is real restoration. SR must score it positively. Yet movement above a poverty threshold does not necessarily establish stable middle-class resilience when social insurance, savings and service access remain weak.

7. Labor, Informality, and Social Protection

Labor informality is the organizing economic fact of the Mexican paper. INEGI placed the national informality rate near 54.5% in 2024, with enormous state variation. Informal work includes heterogeneous activities, from viable enterprises and self-employment to low-paid dependent work without enforceable protection. It cannot be treated as a single pathology.

Informality provides rapid entry, flexibility, local services and household survival. It also commonly limits pensions, occupational protection, paid leave, stable credit, unemployment insurance and direct healthcare entitlement. The SR interpretation is dual: informality is simultaneously adaptive capacity and a channel through which institutional risk becomes human risk.

Low open unemployment can coexist with precarious work because people cannot afford non-employment. The decisive indicators are earnings adequacy, hours, volatility, social insurance, involuntary self-employment, workplace injury and transition probabilities—not unemployment alone.

8. Household Income, Finance, and Remittances

Mexican households combine labor income, informal enterprise, transfers, pensions, family support, assets and remittances. Remittances are a major transnational capacity layer: they stabilize consumption, housing, education, health and local economies, but also expose households to migration policy, U.S. labor conditions, exchange rates and family separation. Their contribution should be measured as real capacity without naturalizing the conditions that make migration necessary.

Financial inclusion has expanded through accounts, mobile services and public payment infrastructure, but cash, informal savings groups, family loans and supplier credit remain important. Formal credit exclusion may reduce over-indebtedness for some households while increasing cost and vulnerability for others. The human-buffer test

monitors liquid savings, borrowing source, remittance dependence, skipped consumption, asset sales and additional work before later service or housing loss.

9. Housing, Water, Food, and Material Survival

Housing adequacy in Mexico cannot be reduced to rent burden. Tenure may involve ownership without clear title, inherited property, self-construction, informal settlement, overcrowding, weak materials, unsafe location or missing services. Incremental construction is a powerful household technology of adaptation; it also transfers finance, labor and safety responsibility to families.

Water is a central cross-domain constraint. Availability, network continuity, quality, extraction, drought, industrial demand and household storage differ sharply by region. Tankers, rooftop tanks, purchased water, household treatment and altered schedules preserve function where networks are intermittent. These are textbook compensatory systems: they create continuity while consuming money and time and concealing network unreliability.

Food security reflects income, prices, local production, transport, water, public programs and household labor. Mexico's official multidimensional framework correctly treats nutritious food access as a right-based deprivation rather than only calorie supply. The forecast monitors dietary substitution, meal reduction and caregiver effort before measurable health and school consequences.

10. Healthcare Capacity and Effective Access

Mexico's health system is segmented across employment-linked social security, services for people without social insurance, state and federal providers, military systems, private care, pharmacies and household provision. Institutional reorganization during the transition from Seguro Popular to INSABI and then IMSS-Bienestar changed financing and delivery while the pandemic altered demand and staffing. Formal affiliation series must therefore be interpreted with institutional breaks.

Effective access depends on medicines, personnel, distance, waiting, referral, language, safety and household payment. Pharmacies with adjacent clinics and private consultations can supply rapid access, but may fragment continuity and shift cost to households. Family members also provide navigation, transport, bedside care and medication management. The correct diagnosis is segmented formal capacity with extensive household and market substitution—not absence of a health system.

Restoration would be visible through reliable primary care, medicine availability, continuity, reduced out-of-pocket burden and declining caregiver effort. Human-buffer inversion would appear first as health-worker exits, family care intensity, delayed treatment and borrowing, followed by preventable complications and local service interruption.

11. Education and Childhood Infrastructure

Mexico has achieved broad educational expansion, but effective learning depends on attendance, teacher presence, language, school infrastructure, digital access, nutrition, safety and household time. Rural, Indigenous, migrant and violence-exposed communities face distinct constraints. A national enrolment rate cannot represent these conversion factors.

Households compensate through transport, devices, tutoring, sibling care, migration decisions and forgone child earnings. During disruption, mothers and older children frequently become the continuity layer. The 2031 registry separates teacher burden, household educational time, youth work and school exit from later learning and attainment outcomes.

12. Care, Gender, and the Household Economy

Unpaid domestic and care work is not peripheral to Mexico's economy. CONEVAL's gender analysis found that women averaged about 32 hours of unpaid household work weekly compared with 14 hours for men, while women in poverty with young children had especially low labor-force participation. This is direct evidence of an unequal human buffer.

Care can embody solidarity, identity and reciprocity while still constraining income, health, education and autonomy. The SR test does not classify family care as exploitation by definition. It asks whether care is chosen, shared, supported and renewable, and whether the provider retains recovery time and access to social protection.

13. Governance and Administrative Burden

Mexico's administrative state has major national capacities in taxation, elections, statistics, monetary policy, social programs and identification, alongside weaker capacity in many municipalities and service interfaces. Digitalization can simplify access, but documentation, travel, queues, interoperability, language and discretion can shift administrative work to users.

Where formal institutions are difficult to navigate, gestores, relatives, community authorities, unions, parties, employers and local brokers may mediate access. Mediation can be helpful or coercive. Its growth is a reorganization signal because function becomes dependent on relational position rather than transparent universal process.

14. Security, Justice, Rights, and Effective Remedy

INEGI registered 33,241 homicides in 2024. This scale is a severe national pressure, but violence is territorially concentrated and heterogeneous. Homicide counts cannot represent extortion, disappearance, displacement, gender violence, police abuse, impunity, fear or community protection. Nor can they establish that every jurisdiction has lost governance.

Security pressure changes ordinary behavior: routes, business hours, schooling, reporting, migration, investment and public participation. Families and firms purchase security, organize collectively, comply with coercive actors, relocate or remain silent. Those responses preserve survival while reducing agency. The institutional threshold is crossed locally when residents cannot safely use essential services or seek remedy and when alternative coercive authority becomes durable.

Formal rights require usable investigation, counsel, courts, protection and enforcement. High impunity or delay shifts evidence collection, search, advocacy and protection to families and civil society. The human-buffer forecast therefore includes search labor, legal cost, journalist and defender exits, and community-security burden before later institutional withdrawal.

15. Demography, Migration, and Recovery Capacity

Mexico remains demographically large and comparatively younger than Canada, but fertility decline, aging and regional depopulation will alter care and labor. Internal and international migration are not merely demographic flows: they redistribute skills, income, caregiving, political voice and risk. Migration can release household pressure while emptying local human capacity.

The national system benefits from diaspora income and knowledge but pays through separation and uneven local reproduction. A municipality can remain financially viable through remittances while losing working-age residents and care capacity. Counts of population and money must therefore be joined to age structure, return migration, school continuity and household care.

16. Digital Infrastructure, Media, and AI

Internet access expanded rapidly, with household connectivity approaching four-fifths by 2025 after roughly doubling from 2015. This is genuine capacity growth. Effective digital agency nevertheless depends on device quality, affordability, skills, language, cybersecurity, identity systems and offline remedy. Mobile access is not equivalent to secure participation in education, banking or government.

Media capacity is strong nationally but dangerous in some local environments. Violence and economic pressure on journalists can create information deserts and self-censorship without eliminating national media. The relevant SR outcome is public legibility: whether people can know what is happening, attribute responsibility and act safely.

AI will enter a system already structured by informal work, uneven records, territorial gaps and mediated access. Automation may extend services and translation; it may also inherit missing data and make informal populations administratively invisible. The 2031 test monitors appeal, human review, false exclusion and frontline workload rather than treating AI adoption as deterioration.

17. Energy, Transport, Industry, and Critical Infrastructure

Mexico's electricity and energy systems combine state enterprises, private participation, fossil resources, growing renewable potential and strong regional differences. Nearshoring increases electricity, water and logistics requirements. Durable industrial gains require transmission, generation, housing, skills, security and environmental legitimacy in the same territories.

Transport combines major ports, highways, freight rail and aviation with household dependence on buses, colectivos, informal transit and long travel. Informal mobility is essential capacity; safety, cost and reliability determine whether jobs and services are effectively accessible. Infrastructure failure often appears first as additional household travel time and worker improvisation.

18. Climate, Ecology, and Disaster Capacity

Mexico faces drought, extreme heat, hurricanes, floods, wildfire, coastal exposure, water stress and ecosystem loss. Exposure is not uniform, and the country possesses substantial civil-protection, engineering, military, community and family response capacity. The forecast concerns repeated shocks that arrive before recovery is complete.

Household storage, informal construction, migration, mutual aid, remittances and unpaid cleanup can restore function after disaster. If recurrence accelerates, those same buffers can become depleted. The predicted order is responder and household burden first, followed by incomplete reconstruction, insurance or credit withdrawal, school interruption, health effects and municipal fiscal stress.

19. Cross-Domain Integration

Mexico's recurring relation is: incomplete or uneven formal reach → household, informal, community and transnational compensation → visible continuity → unequal human burden → possible local compensatory inversion. The system is not simply weak. It possesses multiple suppliers of function, some highly adaptive and legitimate. The danger is that their contribution can make formal deficits difficult to see and politically easy to normalize.

Formal Gap → Relational / Informal Compensation → Continuity → Burden Accumulation → Selective Exhaustion

Dimension	Canada	United States	Mexico
Dominant continuity	Public-system compensation	Stratified purchased substitution	Dual-system informal and relational continuity
Primary filters	Province, staffing, geography	Income, insurance, eligibility, geography	Formality, territory, gender, security, documentation
Human buffer	Workers, households, migration	Debt, work, care, purchase, exit	Unpaid care, informal work, remittances, community, migration
Likely visible failure	Waits, vacancies, delayed public service	Local closure, backlog, withdrawal	Local interruption, under-registration, forced mobility, informal authority
National collapse	Not supported	Not supported	Not supported

19.4 Field-System Architecture

The country model is a field-preserving system. Each domain i has a state vector $x_i(t)$ containing variables appropriate to that domain—such as access, delay, affordability, staffing, reliability, effective remedy, household margin, or ecological exposure. Within-field dynamics are represented as $x_i(t+1) = F_i x_i(t) + G_i z_i(t) + \varepsilon_i(t)$, where z_i contains observed pressures and interventions. The equation is a model class, not an estimated national law. It prevents a finding in one field from silently becoming evidence in another.

A cross-field claim is registered only through an explicit bridge: $x_i(t) \rightarrow M_{i\Box}(t, \tau_{i\Box}) \rightarrow x_{\Box}(t+\tau_{i\Box})$. $M_{i\Box}$ must identify the transmitting exposure or institution, temporal order, transformation, expected lag $\tau_{i\Box}$, amplifiers, buffers, reverse direction, common shocks, and a falsifier. The stacked country system is $X(t+1) = AX(t) + GZ(t) + L\Omega(t) + \varepsilon(t)$. A represents preregistered domestic bridges; Ω represents common, transnational, or planetary shocks. Estimating A is a future empirical task. The present papers register bridge hypotheses and grade the evidence supporting each link.

Pressure Ecology is used as a theorem rather than counted as a twenty-fourth field. It asks how pressure is generated, imported, absorbed, transformed, transferred, externalized, displaced in time, and propagated. For domain k , a bookkeeping identity is $P^*_{\Box}(t+1) = P^*_{\Box}(t) + G_{\Box}(t) + I_{\Box}(t) - A_{\Box}(t) - E_{\Box}(t) - D_{\Box}(t) + \sum_{\Box} T_{\Box\Box} \rightarrow \Box(t) - \sum_{\Box} T_{\Box\Box} \rightarrow \Box(t)$. P^* is residual effective pressure; G is domestic generation; I is imported pressure; A is genuine absorption; E is externalization beyond the boundary; D is temporal displacement; and T is cross-domain transfer. Transformation changes the form of pressure and is recorded in the relevant bridge rather than treated as disappearance. Displaced pressure must be re-entered when its obligation matures.

The field inventory is therefore neither a checklist nor a claim that all twenty-three fields are equally activated. A field is diagnostic in a country paper only when the country evidence identifies its distinct object, measures at least one state variable, specifies a mechanism, defines a prediction, and states a falsifier. "Contextual," "indeterminate," and "not diagnostic" are valid outcomes. The three specialized subfields—the SR Matrix Field, the SR Policy Substrate Field, and Structural Restoration Dynamics—organize integration, intervention, and recovery without erasing the autonomy of the full fields.

The canonical field count used in this study is twenty-three full fields plus three specialized subfields. Pressure Ecology, the Cross-Field Integration Framework, DVS, equations, indices, operators, and diagnostic systems are canonical components but are not counted again as fields.

19.5 Pressure Ecology Ledger

Mexico's pressure ecology is distinguished by conversion between formal and relational systems. Households combine wages, self-employment, transfers, remittances, informal credit, public programs, and unpaid work. Communities and small firms supply logistics, care, water, security, and information where formal provision is incomplete or costly. This is not merely residual coping: it is a durable institutional topology. Its strength is redundancy and local knowledge; its vulnerability is that continuity can be financed through low pay, missing social insurance, migration separation, women's time, legal ambiguity, and exposure to violence.

Process	Country mechanism	Evidentiary judgment
Generation	Informality, territorial inequality, violence, water stress, service gaps, care load, municipal capacity limits	Directly observed in multiple official series.
Importation	U.S. demand and policy, remittance cycles, global manufacturing, energy and food prices, climate	High North American coupling; effects regionally differentiated.
Absorption	Households, remittances, informal work, public transfers, communal systems, formal firms and agencies	Substantial continuity; sustainability varies.
Transformation	Formal access gaps become unpaid work, private purchase, informal service, travel, migration, non-use	Central mechanism; measurement integration weak.
Transfer	Federal/state/municipal; institution → household/community; Mexico ↔ migrant workers abroad	Strong descriptive support; valuation incomplete.
Externalization	Care and income generation shifted across borders; ecological and security costs localized	Essential to interpretation; boundary-sensitive.
Temporal displacement	Deferred care, school exit, water-system maintenance, informal debt, migration separation	Observable proxies; longitudinal follow-up required.
Propagation	Informality–social protection–health; violence–mobility–local economy; climate–water–care	Priority hypotheses; causal maturity mixed.
Restoration	Poverty reduction, health-access improvement, connectivity growth, local capacity and formalization	Real gains coexist with large residual deprivation.

The ledger distinguishes disappearance from relocation. Absorption reduces effective pressure; transformation changes its form; transfer changes its bearer; externalization changes the analytic boundary; temporal displacement changes when it becomes visible. A country can therefore look stable while residual pressure accumulates elsewhere in the coupled system. Conversely, a high level of compensation can represent successful and sustainable institutional flexibility. Classification depends on duration, voluntariness, replacement cost, recovery, and whether the generating gap closes.

19.6 Field Activation Matrix

The matrix records where each canonical field has a distinct country-level object. “Primary” indicates direct relevance and at least one measurable state; “Bridge” indicates that the field is most useful at an interface; “Contextual” indicates that present national evidence does not justify an independent mechanism claim. These labels govern this paper only; they are not judgments about the field's status elsewhere.

No.	Field or subfield	Role	Country object
1	Institutional Cognition	Primary	Visibility, under-registration, recognition and territorial representation.

2	SignalRupture Institutional Governance Field (SR-IGF)	Primary	Federal/state/municipal implementation and burden allocation.
3	Collapse Governance	Bridge	Household, community, informal and migrant compensation.
4	The Drift Collapse Field (DCF)	Primary	Domain thresholds and local functional loss.
5	Systems Propagation Science (SPS)	Primary	Informality–protection–health, violence–mobility, climate–water propagation.
6	SR Cyber-Epistemic Infrastructural Diagnostic Field (SR-CEIDF)	Bridge	Telecom, platforms, cyber, identity and epistemic dependency.
7	The SR Algorithm Field	Bridge	Administrative, credit, labour, security and platform systems.
8	Infrastructural Political Economy	Primary	Informality, industry, labour, remittances, distribution and ownership.
9	Infrastructural Overshoot	Primary	Water, climate, energy, logistics, settlement and ecological exposure.
10	System Inheritance Studies (SIS)	Bridge	Legacy categories and unequal formal infrastructures inherited by digital systems.
11	Collapse Epistemology	Bridge	Media, violence, under-reporting and public legibility.
12	SignalRupture Research Incentive Field (SR-RIF)	Bridge	Research geography, funding, measurement and translation incentives.
13	Collapse Psychology	Primary	Constraint, violence exposure, care load, insecurity and agency.
14	Collapse Sociology	Primary	Household, community, class, gender and territorial stratification.
15	SR Anthropology	Primary	Indigenous/communal authority, migration, kinship, informality and adaptation.
16	Collapse Demography	Primary	Fertility, mortality, migration, household and local recovery.
17	Childhood Cognitive Infrastructure Studies (CCIS)	Primary	School access, nutrition, care, language, violence and connectivity.
18	Infrastructural Medicine	Primary	Formal coverage, effective access, workforce and household substitution.
19	Collapse Criminology	Primary	Violence, impunity, policing, community governance and reporting.
20	Post-Web Jurisprudence	Primary	Effective remedy, evidence, digital access and territorial court capacity.
21	Collapse Religion	Contextual	Contextual: religious and communal provision; independent national mechanism not established.

22	Collapse Economics	Primary	Income, informality, remittances, poverty, household margin and production.
23	Collapse Society	Primary	Country coupling without converting inequality into one collapse score.
24	The SR Matrix Field	Bridge	Cross-domain matrix with state and municipal heterogeneity.
25	The SR Policy Substrate Field	Primary	Policy instruments, implementation substrates and transfer effects.
26	Structural Restoration Dynamics	Primary	Poverty/access gains, burden reduction, reversals and local recovery.

Activation is claim-specific. A “Primary” label does not validate every proposition associated with a field, and a “Contextual” label does not disconfirm the field. The empirical burden remains at the level of construct, mechanism, bridge, population, and prediction.

19.7 Cross-Field Bridge Register

The following bridges are the country model’s priority causal candidates. Each is directional for testing even where feedback is expected. Status describes the present evidence, not inevitability.

Source field	Mechanism/exposure	Receiver	Lag	Status and falsifier
Labour/Infrastructural Political Economy	Informality limits contributory social protection and changes bargaining	Health; household finance; demography	Immediate–20 years	Strong association; sector and selection controls required
Migration/SR Anthropology	Migration converts local labour into remittance and transnational care circuits	Housing; consumption; family; local finance	Months–20 years	Strong flows; causal household effects heterogeneous
Care/Collapse Sociology	Unpaid work compensates for service, income, and time deficits	Women’s labour; health; childhood	Immediate–15 years	Prevalence strong; sustainability threshold unestimated
Water/Overshoot	Scarcity and unreliable service create storage, purchase, travel, illness, and conflict	Health; education; household margin	Immediate–10 years	Local panels and hydrological controls required
Violence/Collapse Criminology	Coercion changes mobility, enterprise, journalism, governance, and reporting	Economics; epistemics; demography	Immediate–10 years	Strong plausibility; reverse causality and undercounting tested
Health/Infrastructural Medicine	Formal coverage and effective access diverge by work and territory	Household care; labour; mortality	Immediate–10 years	Official deprivation measure supports gap; utilization linkage needed
Digital/SR-CEIDF	Connectivity expands access while concentrating platform, telecom, identity, and payment dependencies	Markets; media; administration	Immediate–5 years	Growth confirmed; dependency/failure effect separately tested
Education/CCIS	Material, care, violence, language, and connectivity co-exposures shape persistence	Demography; labour; psychology	1–20 years	Conventional education models first; SR interaction gain open
Governance/SR-IGF	Uneven municipal capacity shifts implementation and navigation to local actors	Rights; services; disaster recovery	Months–10 years	Territorial comparison required; high-capacity municipalities are controls

Climate/Infrastructural Overshoot	Heat, drought, storms, and crop shocks alter water, health, migration, and income	Households; infrastructure; demography	Immediate–20 years	Physical exposure strong; mediation locally tested
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No bridge is supported merely because its source and receiver indicators both deteriorate. The minimum design requires temporal order, a transmitting exposure, an independently measured receiver, common-shock controls, reverse-path testing, and a negative or recovery case. Where a bridge cannot meet this contract, it remains a registered hypothesis rather than part of the empirical conclusion.

19.8 Dependency and Substitution Topology

National capacity depends not only on the quantity of resources but on whether critical nodes can be substituted under stress. For node m , the heuristic dependency relation is $R_m = \kappa_m \chi_m \delta_m / (1 + \sigma_m)$, where κ is functional criticality, χ is provider or pathway concentration, δ is jurisdictional or geographic distance from effective control, and σ is tested substitution capacity. R_m is not estimated here and is not a country score. It states what must be measured before concentration is interpreted as fragility.

Critical function	Dependencies	Substitution capacity	Pressure implication
Household continuity	Women's unpaid work, kinship, informal earnings, shared housing	Low when household health or time is saturated	Core hidden dependency; not infinitely elastic
Remittance circuits	Migrant employment, U.S. economy/policy, transfer rails, exchange rate	Partial through domestic income and public transfers	Transnational buffer with external policy exposure
Social protection	Formal employment, contributory systems, public programs	Unequal for informal workers	Households and private spending substitute for missing coverage
Water	Aquifers, utilities, energy, local governance, household storage/purchase	Highly territorial	Private purchase may preserve use while increasing burden
Manufacturing/logistics	U.S. demand, border crossings, electricity, transport, supplier networks	Moderate in industrial corridors	Integration creates capacity and imported shock exposure
Digital access	Telecom operators, mobile devices, electricity, affordability, platforms	Growing but uneven rural and income substitution	Access expansion coexists with concentrated intermediaries
Security and justice	Police, prosecutors, courts, community authority, journalists and defenders	Low in high-violence or low-capacity localities	Under-reporting can conceal rather than resolve loss of function
Disaster recovery	Municipal capacity, federal/state support, households, migration, remittances	Unequal by territory and tenure	Return and restored service matter more than rebuilt asset totals

Substitution is evaluated by speed, coverage, affordability, quality, legal authority, and reversibility. A substitute that exists only for high-income households or major metropolitan institutions is not national substitution. A local or informal substitute is counted as real capacity, but its labour, health, legal, ecological, and opportunity costs remain part of the system ledger.

19.9 Structural Restoration Dynamics and Negative Cases

Restoration is a change in the generating substrate, not simply a rebound in the surface indicator. The operational condition is $\Delta G < 0$ together with stable or improving function and declining dependence on extraordinary compensation. Compensatory inversion is the contrary warning pattern: $G > 0$, $\Delta C_e \leq 0$, and subsequent $\Delta F < 0$, where G is the requirement–capacity gap, C_e extraordinary compensation, and F observed function. The present evidence does not establish a national compensatory inversion.

Domain	Countertrend or restoration evidence	Interpretation
Poverty	Multidimensional poverty fell from 43.2% in 2016 to 29.6% in 2024	Major national improvement; rural poverty remained 45.8% and territorial inequality persists.

Extreme poverty	Population in extreme poverty fell from 9.1 million in 2022 to 7.0 million in 2024	A real countertrend; do not reclassify it as hidden collapse.
Health access	Deprivation in access to health services fell from 39.1% to 34.2% between 2022 and 2024	Improvement is substantial; 44.5 million people still experienced deprivation.
Social security	Deprivation declined to 48.2% in 2024	Direction improved while the remaining level shows continuing dual-system dependence.
Digital access	Internet use reached 83.1% in 2024 and continued rising	Restoration/expansion with an urban–rural gap: 86.9% versus 68.5% in 2024.
Formal industry	Large export and manufacturing capacity persists	Rejects low-capacity generalization; dependency on North American cycles remains.
Community systems	Communal and household infrastructures repeatedly preserve function	Count as capacity, while measuring whether the burden is voluntary, equitable, and sustainable.
Transfers	Official analysis shows transfers materially reduce measured poverty	A policy mechanism with measurable effect; durability and substrate change remain separate.

Negative cases are part of the test rather than rhetorical balance. A valid national model must explain why some domains improve, why some exposed places avoid deterioration, why some policies reduce burden, and why the same pressure does not produce the same result everywhere. These cases identify buffers, threshold differences, institutional learning, and omitted variables. If the model cannot accommodate them without changing definitions after observation, the integrated claim fails.

19.10 DVS Validation and Robustness Tests

The Diagnostic Validation Standard (DVS) separates the existence of a field, the evidence for a country claim, and institutional recognition. The country model proceeds through operationalization, measurement validation, longitudinal or comparative testing, strong conventional controls, mechanism testing, adversarial falsification, replication, and intervention or restoration testing. The following checks govern interpretation of this manuscript.

Check	Application	Status
Poverty source validation	INEGI 2024 multidimensional poverty: 29.6%, 38.5 million; 2016–2024 decline from 43.2%.	Confirmed
Access disaggregation	Health-service deprivation 34.2% and social-security deprivation 48.2% in 2024 kept separate.	Confirmed
Labour validation	ENOE fourth-quarter 2024 labour informality approximately 54.3%; not treated as collapse by definition.	Confirmed
Digital validation	ENDUTIH 2024: 83.1% internet use; urban 86.9%, rural 68.5%.	Confirmed
Violence validation	INEGI registered 33,241 homicides in 2024; under-registration and territorial concentration remain model concerns.	Confirmed
Negative cases	Poverty, health-access, social-security, connectivity, industrial and transfer gains retained.	Applied
Conventional baselines	Labour informality, development, federalism, violence, migration, hydrology, and social-policy models estimated first.	Required
Forecast lock	2031 test requires measured buffer saturation before local functional deterioration, with restoration pathways scored equally.	Falsifiable

Incremental value is the decisive comparison. For each outcome y, the strongest mature disciplinary model B is estimated first; SR variables and registered bridges are added only afterward. The claim survives only if B+SR

improves held-out prediction, calibration, warning lead time, mechanism discrimination, or intervention choice. In notation: $\Delta V = V(B+SR) - V(B)$. A non-positive, unstable, or non-replicating ΔV weakens the relevant SR claim even when the descriptive synthesis remains useful.

Robustness requires alternative denominators, thresholds, lags, regional units, population weights, missing-data assumptions, and bridge directions. Results must be reported with confidence or uncertainty intervals where estimable. Post-hoc index weights are prohibited. Measurement changes and publication lags are recorded as possible visibility effects but cannot be treated as proof of institutional concealment.

19.11 Integrated Country Judgment

Mexico's pressure ecology is distinguished by conversion between formal and relational systems. Households combine wages, self-employment, transfers, remittances, informal credit, public programs, and unpaid work. Communities and small firms supply logistics, care, water, security, and information where formal provision is incomplete or costly. This is not merely residual coping: it is a durable institutional topology. Its strength is redundancy and local knowledge; its vulnerability is that continuity can be financed through low pay, missing social insurance, migration separation, women's time, legal ambiguity, and exposure to violence.

The field system adds value at the level of disciplined integration: it makes burden bearers, substitutions, dependencies, temporal displacement, and recovery conditions explicit across domains. It does not replace the mature literatures that identify domain mechanisms, and it does not convert simultaneous pressures into one causal story. The national finding is therefore a bounded topology of continuity and exposure. Its strongest causal and predictive claims remain registered for panel, quasi-experimental, and held-out testing.

19.12 AI Substrate Transition: CRI, AIS, and MACS

The AI environment is modeled as a substrate transition rather than a standalone technology sector. Computational recognition infrastructure (CRI) detects and classifies patterns. AI-mediated institutional substrates (AIS) place those systems inside eligibility, triage, pricing, enforcement, forecasting, allocation, procurement, scheduling, and remedy. Multi-agent coordination systems (MACS) arise when interacting automated agents operate across institutions, firms, households, platforms, and countries. Each phase can increase capacity while also inheriting the country's existing categories, dependencies, inequalities, and continuity mechanisms.

Mexico is likely to adopt AI across a formal–informal dual system. Mobile translation, education, remittance navigation, informal commerce, health triage, local logistics, and public administration could broaden effective capacity. The same systems may reproduce incomplete records, language and identity exclusions, territorial connectivity gaps, concentrated foreign infrastructure, and limited appeal. The decisive question is whether AI increases local agency and domestic value retention or deepens external technological dependency.

Phase	Country insertion	Registered prediction	Decisive test
CRI	Pattern recognition in administration, remittances, commerce, health, education, security, and logistics	Low-cost navigation and earlier local visibility	Rural, Indigenous, informal, and low-connectivity populations gain measurable access
AIS	AI embedded in eligibility, identity, credit, labour, border, platform, and municipal systems	Formal efficiency may coexist with exclusion from incomplete records	Error, exclusion, appeal, and language outcomes improve across territories
MACS	Interacting public, platform, manufacturing, logistics, migrant, and household agents	Coordination improves while foreign cloud/model dependency concentrates	Local substitution, data governance, and service continuity survive provider disruption
2031 test	Mobile capacity expands faster than sovereign infrastructure and institutional remedy	AI either raises local agency or transmits more value and control outward	Domestic capability, value retention, workload, access, and bargaining power all improve

The five-year prediction is directional and conditional. AI should initially extend human and organizational buffers by accelerating documentation, triage, forecasting, and coordination. Surface throughput may therefore improve before effective access, household margin, workload, rights, or outcomes. As delegation deepens, cloud and model providers become critical dependency nodes, institutional visibility may lag behind operational change, and human-buffer exhaustion may reappear as exception handling, monitoring, appeals, and responsibility without authority. The AI claim is supported only if these sequences outperform conventional technology-adoption, labour, organizational, and inequality models in held-out tests.

AI counts as structural restoration only when it reduces the generating requirement–capacity gap while maintaining or improving function and lowering extraordinary workload, exclusion, dependency, and transferred burden. If AI improves headline throughput while those substrate measures worsen, it is classified as compensation, temporal displacement, or pressure transfer—not restoration.

19.12.1 Locked Forecast Ladder, 2026–2031

The 2031 endpoint is retained as the final adjudication date, but it is not the first evaluation date. The following checkpoints are registered prospectively so an acceleration during 2028–2029 can be detected without moving the forecast after observation. Each checkpoint tests a stage in the predicted sequence rather than requiring a national crisis or uniform outcome.

Checkpoint	Model stage	Country observation	Decision rule
2026–2027	Baseline formation	Record formal and informal AI use, connectivity, language and identity access, administrative workload, local ownership, foreign infrastructure dependence, domestic value retention, and territorial variation.	Baseline is frozen before later agency or dependency outcomes are interpreted.
2028	Early divergence	Mobile and low-cost AI capacity should expand before equivalent gains appear in institutional remedy, complete records, rural and Indigenous access, or locally governed infrastructure.	Broad territorial access and remedy gains weaken the dual-system divergence prediction.
2029	Acceleration test	Foreign cloud/model dependence, record exclusion, language barriers, exception work, and outward transmission of data, rents, or control should become more visible if adoption outruns sovereign capacity.	Comparable adoption with stronger local ownership and value retention weakens the dependency sequence.
2030	Propagation test	AI-mediated effects should cross informal commerce, remittances, administration, manufacturing, logistics, health, education, migration, and municipal capacity.	No reproducible territorial or cross-domain order after connectivity and development controls rejects propagation.
2031	Adjudication	Classify the observed trajectory as locally governed restoration, dual-system compensation, or intensified external technological dependency.	Restoration requires domestic capability, value retention, access, workload, remedy, substitution, and bargaining capacity to improve together.

Temporal compression is recorded when the registered sequence appears earlier than expected. Earlier timing does not validate later stages automatically; each mechanism, bridge, distributional effect, and restoration test must still be observed independently. Conversely, failure of the 2028 or 2029 leading indicators under substantial adoption weakens the forecast before 2031. The model therefore cannot preserve itself by postponing evaluation to the final year.

19.13 International Pressure Externalization and Developmental Lock-In

The international ledger records a directional vector $\Omega^d_{c \rightarrow r}(t)$, where c is the sending country, r the receiving country, and d the domain. It is multidimensional and cannot be collapsed into a single externalization score. Every registered pathway must identify the origin, mechanism, receiver, benefit bearer, cost bearer, lag, substitution or refusal capacity, return pathway, evidence, and falsifier. Ordinary interdependence is not equivalent to asymmetric pressure transfer, and asymmetric transfer is not automatically developmental destabilization.

Mexico is modeled as receiver, absorber, and transmitter. It receives strong U.S.-centered demand, monetary, migration, platform, security, and technological pressure; exports manufacturing, agricultural, logistics, labour, and migrant capacity northward; and can transmit migration-control and return pressure toward Guatemala, Honduras, El Salvador, Haiti, Cuba, Venezuela, and Nicaragua. The model must therefore distinguish pressure generated in Mexico from pressure routed through Mexican institutions under wider North American arrangements.

Receiver or corridor	Pressure pathway	Benefit and cost incidence	Evidence and falsification test
United States; Canada	Manufacturing, agriculture, logistics, labour migration, remittances, trade demand, and nearshoring	North American firms and consumers gain capacity; Mexican workers and regions may gain income while bearing labour, water, housing, and infrastructure loads	Test domestic value retained, wages, supplier upgrading, public revenue, ecological repair, and technological autonomy
Guatemala; Honduras; El Salvador	Transit containment, returns, border enforcement, labour and family networks	Mexico and the United States gain migration-management capacity; migrants and origin communities bear variable interruption and reintegration costs	Trace decision authority, return volumes, safety, reintegration, remittance loss, and origin-system recovery
Haiti; Cuba; Venezuela; Nicaragua	Asylum, transit, temporary settlement, enforcement, and return arrangements	Administrative containment may transfer housing, care, legal, and security pressure across borders and municipalities	Disaggregate nationality, legal status, duration, remedy, municipal burden, onward movement, and safe return
Foreign cloud, model, platform, and industrial providers	AI infrastructure, data governance, identity, logistics, finance, and production control	Local users gain capability while rents, standards, data control, and common-mode risk may remain external	Measure local ownership, skills transfer, data authority, switching capacity, domestic suppliers, and service continuity

Developmental destabilization is registered only where the pathway persistently weakens a receiving country's fiscal, institutional, ecological, labour-reproductive, technological, or recovery capacity. The developmental-lock relation is $L_r(t+1) = f[X_r, H_r, D_r, E_r, C_r] - g[V_r, S_r, R_r, B_r]$. X is externally captured value; H human-capacity drain; D debt or foreign-exchange exposure; E ecological residual load; C institutional capture or constraint; V domestic value retention; S substitution; R restoration; and B bargaining and regulatory capacity. This is a test architecture, not an estimated index.

The hypothesized lock sequence is: external extraction or policy pressure → reduced domestic value retention → ecological, labour, fiscal, or institutional burden → weakened recovery and substitution → increased dependence on foreign capital, markets, technology, or migration → reduced bargaining capacity → renewed externalization. Confirming evidence would include high extraction or labour export with weak value retention, loss of trained capacity, unrepaired ecological liabilities, foreign-currency or platform dependence, stalled structural transformation, weak fiscal or regulatory capacity, and repeated recovery failure. The hypothesis weakens when similarly exposed countries improve productivity, diversification, public revenue, local infrastructure, labour conditions, ecological restoration, substitution, and policy autonomy.

AI adds a new externalization channel. Model ownership, cloud control, rents, and standards may concentrate in one jurisdiction while electricity, water, minerals, data work, content moderation, and downstream administrative risk are distributed elsewhere. By 2031, the registered prediction is that AI will increase both cross-border coordination and common-mode dependency. It will contribute to developmental lock-in only where receiving countries lose value retention, substitution, remedy, or bargaining capacity; diffusion that builds sovereign or locally governed capability is a direct negative case.

19.14 Canonical Planetary Country Protocol

Every Planetary SR country paper uses the same fixed analytic sequence so national differences are comparable without forcing them into a single rank or collapse score.

Stage	Required module	Minimum country-paper content
1–3	Boundary, baseline, and surface/substrate separation	Define country system, period, populations, mature disciplinary baselines, and the difference between output and lived or institutional capacity.
4–6	Field activation, pressure ecology, and human buffer	Activate only diagnostic fields; trace generation, import, absorption, transformation, transfer, externalization, displacement, propagation, and compensation.

7–9	Bridges, dependencies, and unequal exposure	Register directional cross-field mechanisms, critical nodes, substitution, geography, institutions, households, firms, communities, and populations.
10–11	Restoration and negative cases	Distinguish substrate repair from rebound or compensation; retain improvements, resistant cases, and failed predictions.
12	AI substrate transition	Assess CRI, AIS, and MACS; inheritance, workload, access, contestability, providers, common-mode failure, and 2031 predictions.
13	International pressure ledger	Name sender, receiver, mechanism, benefit bearer, cost bearer, lag, refusal, return path, evidence, and falsifier.
14	Developmental lock-in test	Test value retention, human-capacity drain, debt/FX exposure, ecological residuals, institutional constraint, restoration, substitution, and bargaining.
15	Validation and forecast lock	Estimate conventional baselines first; test incremental value, sensitivity, causal order, uncertainty, preregistered predictions, and falsification.

The protocol prohibits deterministic country labels. No country is classified as doomed, permanently developing, collapsed, or causally responsible for another country's condition from exposure data alone. Conclusions remain pathway-specific, time-bounded, distribution-sensitive, and revisable only through stated evidence and falsifiers.

20. National Empirical Scorecard

Proposition	Evidence	Judgment
Dual-system continuity	Informal work, unpaid care, remittances, private and community provision are large and durable.	Supported structurally
Surface–substrate divergence	Poverty reduction and connectivity gains coexist with high informality, access gaps and violence.	Supported
Human-buffer precedence	Mechanism and indicators specified; national temporal test not yet completed.	Prospective / unvalidated
Territorial propagation	Cross-domain pathways are strong candidates but require municipal panels.	Partial
Reorganization	Health, digital, migration and informal governance show supplier changes.	Supported; effects mixed
Incremental predictive value	No held-out comparison completed.	Not achieved
National collapse	Capacity, restoration and heterogeneity contradict a unified threshold.	Not supported

21. Alternative Explanations

Poverty, labor informality, state capacity, gender inequality, organized violence, migration, federalism, social determinants of health and climate-vulnerability literatures may explain the evidence without SR. Planetary SR earns value only if the frozen architecture improves temporal integration, early warning or intervention choice beyond those mature approaches. Re-description is not validation.

22. Limitations

- National averages conceal extreme municipal and state variation.
- Series breaks in health institutions complicate temporal comparison.
- Informality is heterogeneous and cannot be scored uniformly.
- Different surveys do not observe the same households.
- Violence and displacement are systematically under-observed in some settings.
- Indigenous and community institutions cannot be represented adequately by a state–market binary.
- Human Recovery Slack is operationalized by indicators but not falsely calibrated as a national scalar.
- No causal or out-of-sample predictive test has yet validated the full sequence.

23. Five-Year Forecast: Human Buffer Precedence

Forecast horizon: 24 August 2026 to 24 August 2031. Candidate human-substrate crisis window: 2031–2034, only if basic-needs and access burdens remain elevated through 2027–2029, compensatory capacity stops expanding, and Human Recovery Slack fails to recover. This is a timestamped conditional warning, not a declaration of inevitable collapse.

$$G_{\square} = Q_{\square} - (D_{\square} + C_{\square, \square})$$

$G_{\square} > 0$ and $\Delta H_{\square} > 0 \Rightarrow$ visible function may remain stable; $G_{\square} > 0$ and $\Delta H_{\square} \leq 0 \Rightarrow$ later functional loss becomes more likely.

Domain	Human-buffer signal expected first	Later function signal	2031 base prediction
Labor / protection	Multiple jobs, unstable hours, illness without leave, informal persistence	Exit, poverty reversal, pension and care gaps	Unequal saturation
Care / gender	Unpaid hours, reduced employment, caregiver health	Unmet care, school and health discontinuity	Women lead measured burden
Health	Family payment, medicine search, staff turnover, delayed care	Preventable complication and local interruption	Local inversion
Water / housing	Storage, purchased water, self-construction, relocation	Network failure, displacement, unsafe housing	Climate intensifies burden
Security / justice	Protective spending, silence, search and advocacy labor	Service avoidance, forced mobility, informal coercive authority	Territorial inversion
Education	Parent/sibling time, youth work, teacher load	School exit and learning loss	Regional divergence
Municipal / disaster	Volunteer and responder load, remittance rebuilding	Incomplete restoration and fiscal incapacity	Repeated shocks expose limits

23.1 Evaluation and falsification rule

Support requires at least two pre-specified human-buffer indicators to deteriorate before or during later institutional deterioration in at least three domains and in replicated territorial panels. Strong support additionally requires held-out predictive gain over conventional baselines. The forecast fails if institutional function generally deteriorates first, if human buffers replenish while function worsens, or if durable capacity expansion reduces burden and failure without the predicted inversion.

23.2 Competing pathways

Pathway	Observable pattern	Interpretation
Restoration	Formal employment, social protection, water, health, safety and municipal capacity expand; burden falls.	Exhaustion forecast does not occur
Renewable dual continuity	Informal and community systems remain voluntary, effective and renewable.	Resilience without inversion
Selective inversion	Exposed households and territories lose time, health, income and relational capacity before service loss.	Base case
Direct shock	Violence, fiscal, climate or infrastructure event destroys function before measured buffer decline.	Precedence falsified for episode
Synchronized national failure	Multiple domains cross essential-function thresholds across most regions.	Low-probability, not base case

24. Decisive Next Test

Construct a preregistered municipality-year and state-year panel, with household microdata where lawful, covering income, informality, social insurance, unpaid time, remittances, water continuity, health access, school attendance, violence, displacement, infrastructure and disaster recovery. Freeze outcomes, lags, subgroup rules and holdout territories before estimation.

1. Estimate mature domain baselines first.
2. Separate formal coverage from effective access.
3. Measure community and informal capacity without assuming it is either failure or virtue.
4. Test whether compensatory indicators buffer current function and predict later burden.
5. Use policy transitions and credible natural experiments where assumptions hold.
6. Hold out states, municipalities and periods.
7. Score restoration, nulls and reversals identically to deterioration.

25. What Mexico Establishes

Mexico establishes that the Planetary SR human buffer cannot be equated with private purchase or public-sector overtime. It can be an entire relational economy: informal work, unpaid care, remittances, self-construction, local transport, family navigation, community authority and migration. These systems are neither statistical residue nor automatically pathological. They are real infrastructure.

The finding also sharpens the theory. Human-buffer exhaustion may begin from a lower measured level of formal slack while remaining culturally and administratively normalized for longer. Visible institutional deterioration can therefore be delayed not by abundant money but by flexible labor, kinship, migration and territorial adaptation. The ethical and empirical task is to distinguish renewable reciprocity from coerced substitution.

26. Final Empirical Judgment

Mexico is not classified as a collapse state. It is a high-capacity, unequal, territorially heterogeneous country whose continuity is jointly produced by formal institutions and extensive informal, household, community and transnational systems. Poverty reduction, connectivity expansion, industrial opportunity and demographic capacity are real restorations. High informality, social-protection gaps, violence, water stress and care inequality remain serious pressures.

The strongest valid diagnosis is dual-system continuity under territorial inequality. The strongest prediction is selective human-buffer saturation before later local functional deterioration. The claim is prospective and falsifiable, not validated by the descriptive evidence alone.

27. Conclusion

Canada, the United States and Mexico now reveal three distinct national conversion mechanisms under one frozen grammar. Canada relies more heavily on compensation beneath universal public structures. The United States filters high aggregate capacity through purchasing power and eligibility. Mexico combines formal institutions with informal and relational infrastructures that are indispensable to continuity.

The North American result is therefore not a common collapse score. It is a comparative topology of who supplies function when durable institutional capacity is insufficient. Planetary SR survives the Mexico test provisionally because it preserves improvement, heterogeneity, alternative explanations and falsification while producing a diagnosis that neither Canada nor the United States could substitute.

References

Banco de México. Remittances and financial-system statistical releases, 2024–2026.

Consejo Nacional de Evaluación de la Política de Desarrollo Social (CONEVAL). Medición multidimensional de la pobreza 2022; Informe de pobreza multidimensional 2022 (2025).

CONEVAL. Sistema de Indicadores sobre Pobreza y Género, 2016–2022.

Instituto Nacional de Estadística y Geografía (INEGI). Encuesta Nacional de Ocupación y Empleo, 2024–2026.

INEGI. Encuesta Nacional sobre Uso del Tiempo 2019 and 2024.

INEGI. Encuesta Nacional para el Sistema de Cuidados 2022.

INEGI. ENDUTIH 2024–2025: availability and use of information technologies in households.

INEGI. Defunciones por homicidio, preliminary and final releases for 2024.

Instituto Nacional de Salud Pública. Encuesta Nacional de Salud y Nutrición, continuous releases.

World Bank. Mexico Overview and Mexico Poverty and Equity assessments, 2024–2025.

Additional Primary Sources

Instituto Nacional de Estadística y Geografía. (2025). Medición de la pobreza multidimensional, 2024.
https://www.inegi.org.mx/contenidos/saladeprensa/boletines/2025/pm/pm2025_08.pdf

Instituto Nacional de Estadística y Geografía. (2025). Encuesta Nacional sobre Disponibilidad y Uso de Tecnologías de la Información en los Hogares (ENDUTIH) 2024.
https://www.inegi.org.mx/contenidos/saladeprensa/boletines/2025/endutih/ENDUTIH_24_RR.pdf

Instituto Nacional de Estadística y Geografía. (2025). Defunciones registradas por homicidio, 2024: resultados preliminares. <https://en.www.inegi.org.mx/app/saladeprensa/noticia/10077>

Banco de México. (2026). Workers' remittances, Economic Information System.
<https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadroAnalitico&idCuadro=CA11&locale=en§or=1>

Instituto Federal de Telecomunicaciones. (2024). Diagnóstico del estado que guarda el Internet en México 2024.
<https://www.ift.org.mx/estadisticas/diagnostico-del-estado-que-guarda-el-internet-en-mexico-2024>

International Organization for Migration. (2026). Most returnees see their future in Guatemala: Tailored support needed. <https://www.iom.int/news/most-returnees-see-their-future-guatemala-tailored-support-new-iom-mpi-report>

Cámara de Diputados. (2024). Migración, capital humano y remesas en México.
<https://www.diputados.gob.mx/sedia/sia/se/SAE-ASS-01-24.pdf>

Cámara de Diputados. (2026). Gaceta Parlamentaria: migración y asilo.
<https://gaceta.diputados.gob.mx/Gaceta/66/2026/jul/20260706.html>

Appendix A — Construct Registry

Domain	Pressure	Capacity / compensation	Outcome
Labor	Income and employment requirements	Formal jobs, informal work, enterprise, family support	Adequacy, protection, volatility

Domain	Pressure	Capacity / compensation	Outcome
Care	Children, disability, illness, aging	Services, unpaid care, kinship	Care completion and provider burden
Health	Disease and access requirement	Public/social-security/private care, pharmacies, family	Timely effective care
Water / housing	Shelter and reliable services	Networks, storage, tankers, self-construction	Continuity, safety, displacement
Security / justice	Violence and remedy requirement	Police, courts, family, community, private security	Safety, reporting, remedy
Education	Learning and attendance	Schools, teachers, households, digital tools	Learning, progression, exit
Digital / AI	Connectivity and decision access	Networks, devices, intermediaries, human review	Usable service and contestability
Climate	Hazard and restoration load	Civil protection, communities, remittances, migration	Recovery time and continuity

Appendix B — Frozen Replication Rules

- Preserve the 2026 baseline unchanged; later evidence is appended and scored.
- Prioritize within-country trajectories before cross-country levels.
- Report missing and non-comparable measures rather than favorable substitutes.
- Keep Human Recovery Slack, CL_N and CL_E operationalized but not falsely calibrated.
- Do not merge domains into a scalar national-collapse score.
- Do not classify informal or community provision as failure without measuring function, choice, burden and renewal.
- Require timestamped prediction, outcome window, falsifiers and auditable judgment.

Appendix C — 2031 Human Buffer Registry

Registry date: 24 August 2026. Formal evaluation after sufficient 2027–2028 evidence. Outcome window closes 24 August 2031 for the five-year test; 2031–2034 remains only a candidate crisis window under the stated preconditions.

ID	Domain	Buffer proxy	Later function proxy	Expected order
MX-HB-01	Labor	Informal persistence, extra work, leave loss, savings	Exit, poverty reversal, protection loss	Buffer first
MX-HB-02	Care	Unpaid hours, provider health, employment reduction	Unmet care and dependent outcomes	Buffer first
MX-HB-03	Health	Payment, medicine search, staff exit, delay	Preventable complication, interruption	Buffer first
MX-HB-04	Water/housing	Purchased water, storage, self-build labor, moves	Network loss, unsafe housing, displacement	Buffer first
MX-HB-05	Security/justice	Protection, silence, search and legal labor	Avoidance, forced mobility, remedy loss	Buffer first

ID	Domain	Buffer proxy	Later function proxy	Expected order
MX-HB-06	Education	Household time, youth work, teacher load	Absence, exit, learning loss	Buffer first
MX-HB-07	Disaster/municipal	Responder load, remittance rebuilding	Incomplete restoration, fiscal failure	Buffer first